Problem 1. Let X be a continuous random variable with PDF

$$f_X(x) = \begin{cases} x^2(2x + \frac{3}{2}) & 0 < x \leq 1 \\ 0 & \text{otherwise} \end{cases}$$

If Y=2X+3, find Var(Y).

$$Var(Y) = Var(2X + 3) = 4Var(X)$$

Hence, we should find variance of X first. Hence,

$$Var(X) = E[X^2] - (E[X])^2$$
$$E[X] = \int_{-\infty}^{\infty} x f_X(x)\, dx = \int_0^1 x x^2 \left(2x + \frac{3}{2}\right) dx$$

$$E[X] = \int_0^1 x^3 \left(2x + \frac{3}{2}\right) dx = \int_0^1 2x^4 + \frac{3}{2}x^3 dx = \frac{2x^5}{5} + \frac{3x^4}{8}\bigg|_0^1$$

$$E[X] = \frac{2}{5} + \frac{3}{8} = \frac{16+15}{40} = \frac{31}{40}$$

$$E[X^2] = \int_{-\infty}^{\infty} x^2 f_X(x) = \int_0^1 x^4 \left(2x + \frac{3}{2}\right) dx = \int_0^1 2x^5 + \frac{3}{2}x^4 dx = \frac{2x^6}{6} + \frac{3}{2}\frac{x^5}{5}\bigg|_0^1$$

$$E[X^2] = \frac{2}{6} + \frac{3}{10} = \frac{10+9}{30} = \frac{19}{30}$$

$$Var(X) = E[X^2] - (E[X])^2 = \frac{19}{30} - \left(\frac{31}{40}\right)^2 = \frac{19}{30} - \frac{31}{40} \cdot \frac{31}{40}$$

$$= \frac{3040 - 2883}{4800} = \frac{157}{4800}$$

$$Var(Y) = 4Var(X) = \frac{157}{1200} = 0.13083$$

Problem 2. Let X be a random variable with PDF $f_X(x)$. Find the PDF of the random variable $Y = |X|$

$$f_X(x) = \begin{cases} \frac{1}{3} & -2 < x \leq 1 \\ 0 & \text{otherwise} \end{cases}$$

we use the CDF method.

$$F_Y(y) = P(Y \leq y) = P(|X| \leq y) = P(-y \leq X \leq y)$$

$$F_Y(y) = F_X(y) - F_X(-y)$$

Take derivative with respect to y from both sides

$$f_Y(y) = f_X(y) + f_X(-y);$$ where y is positive

if $0 < y < 1$, then

$$f_Y(y) = \frac{1}{3} + \frac{1}{3} = \frac{2}{3}$$

if $1 < y < 2$, then

$$f_Y(y) = \underbrace{f_X(y)}_{0} + \underbrace{f_X(-y)}_{\frac{1}{3}} = \frac{1}{3}$$

if $y > 2$, then

$$f_Y(y) = \underbrace{f_X(y)}_{0} + \underbrace{f_X(-y)}_{0} = 0$$

$\Rightarrow$

$$f_Y(y) = \begin{cases} \frac{2}{3} & 0 < y \leq 1 \\ \frac{1}{3} & 1 < y < 2 \\ 0 & \text{otherwise} \end{cases}$$

Problem 3. Let $Z \sim N(0,1)$. Find $P(0.5 < |Z - 12| < 1.5)$.

$$0.5 < |Z-12| < 1.5 \iff 0.5 < Z-12 < 1.5 \quad \text{or} \quad -1.5 < Z-12 < -0.5$$

$$\iff 12.5 < Z < 13.5 \quad \text{or} \quad 10.5 < Z < 11.5$$

$$P(0.5 < |Z-12| < 1.5) = P(12.5 < Z < 13.5) + P(10.5 < Z < 11.5)$$

$$= \Phi(13.5) - \Phi(12.5) + \Phi(11.5) - \Phi(10.5)$$

where Φ is the CDF of standard normal Z.

Problem 4. In this problem you may neglect the probability of twins.

(a) A family has 5 children. Find the probability that they have 2 boys and 3 girls.

(b) A family decides to have children until they have 3 girls. Find the probability that they have 2 boys.

(c) A family decides to have children until they have 3 girls. Let C be the total number of children in the family. Compute E[C] and V(C).

(a) In general for each child there are two options boy or girl. Hence If the total number of children is n, then X = number of boys.

$$X \sim Bin(n, \tfrac{1}{2})$$

$$P(X=2) = \binom{5}{2}\left(\frac{1}{2}\right)^5 = \frac{10}{2^5} = \frac{5}{16}$$

(b) number of trials until they have three girls is follows a negative binomial distribution with $r=3$, $p=\frac{1}{2}$.

The experiment stops when they have 3 girls, so having 2 boys in this experiment means in total they have 2 boys + 3 girls = 5.

If X = number of trials until having 3 girls.

$$X \sim NBin(r=3, p=\tfrac{1}{2})$$

Hence the question is asking for $P(X=5)$ =

$$P(X=5) = \binom{4}{2}\left(\frac{1}{2}\right)^5 = \frac{6}{32} = \frac{3}{16}$$

(c) If X is the number of trials until they have 3 girls. $X \sim NBin(r=3, p=\frac{1}{2})$

If you note in each trial a child is added to the number of children.

$$P(C=k) = P(X=k)$$, Hence $E[C] = E[X]$

$Var(C) = Var(C)$

$$E[C] = r \cdot \frac{1}{p} = 6$$

$$Var(C) = r\,\frac{1-p}{p^2} = 3\left(\frac{1}{2}\right) 4 = 6$$

Problem 5. 60 scientists from 30 universities attend a conference. The conference includes a lunch in a cafeteria which has 30 tables each suitable for 2 people. The people seated for lunch at random. Let $X_j = 1$ if the scientists from university j seat together and $X_j = 0$ otherwise.

(a) Compute $E(X_1)$ and $V(X_1)$.

(b) Compute $Cov(X_1, X_2)$.

(c) Let $X = X_1 + X_2 + \cdots + X_{30}$ be the number of tables occupied by the people from the same university. Compute E[X] and V(X).

$$X_j = \begin{cases} 1 & \text{if scientists from university } j \text{ seat together} \\ 0 & \text{otherwise.} \end{cases}$$

$$p(\text{seating together}) = \frac{\boxed{60}\ \boxed{1}}{\boxed{60}\ \boxed{59}} = \frac{1}{59}$$

$$X_j \sim Bern(p) \text{ where } p = P(X_j = 1) = p(\text{seating together})$$
$$= \frac{1}{59}.$$

$$E[X_j] = \frac{1}{59}$$

$$\operatorname{var}(X_j) = \frac{1}{59}\left(1 - \frac{1}{59}\right) = \frac{58}{(59)^2}$$

$$\operatorname{Cov}(X_1, X_2) = E[X_1 X_2] - E[X_1]\,E[X_2]$$

$$E[X_1 X_2] = p(X_1 = 1 \;\&\; X_2 = 1)$$

$$= p(X_2 = 1 \mid X_1 = 1)\; p(X_1 = 1)$$

$$E[X_1 X_2] = \frac{58 \quad 1}{58 \cdot 57} \quad \frac{60 \quad 1}{60 \cdot 59} = \frac{1}{57} \cdot \frac{1}{59}$$

$$\operatorname{Cov}(X_1, X_2) = \frac{1}{57} \cdot \frac{1}{59} - \frac{1}{59} \cdot \frac{1}{59} = \frac{2}{(59)^2\, 57}$$

generally $\operatorname{cov}(X_i, X_j) = \dfrac{2}{(59)^2\, 57}$ when $i \neq j$

$$\operatorname{Cov}\left(\sum_{i=1}^{30} X_i, \sum_{i=1}^{30} X_i\right) = \sum_{i=1}^{30} \operatorname{var}(X_i) + 2 \sum_{i<j} \operatorname{Cov}(X_i, X_j)$$

$$\operatorname{Cov}\left(\sum_{i=1}^{30} X_i, \sum_{i=1}^{30} X_i\right) = \sum_{i=1}^{30} \operatorname{var}(X_i) + 2 \binom{30}{2} \frac{2}{(59)^2\, 57}$$

$$= 30 \frac{58}{(59)^2} + \frac{30 \cdot 29 \cdot 2}{(59)^2\, 57}$$

$$= \frac{30}{(59)^2} \cdot 58 \left[1 + \frac{1}{57} \right] = \frac{30}{57} \left(\frac{58}{59}\right)^2$$

By linearity of Expectation,

$$E\left[\sum_{i=1}^{30} X_i\right] = \sum_{i=1}^{30} E[X_i] = 30 \, E[X_1] = \frac{30}{59}$$

Problem 6. Let X be the number of Heads in 10 fair coin tosses.

(a) Find the conditional PMF of X, given that the first two tosses both land Heads.

(b) Find the conditional PMF of X, given that at least two tosses land Heads.

$$P(X=k \mid H_1H_2) = \frac{P(X=k \;\&\; H_1H_2)}{P(H_1H_2)} = \frac{\binom{10-2}{k-2}\left(\frac{1}{2}\right)^{10}}{\frac{1}{4}}$$

$$P(X=k \mid H_1H_2) = \binom{10-2}{k-2}\left(\frac{1}{2}\right)^{10-2} \quad ; \quad k=2,3,\ldots,10$$

$$P(X=k \mid H_1H_2) = \binom{8}{k-2}\left(\frac{1}{2}\right)^{8} \quad ; \quad k=2,3,\ldots,10$$

At least two heads = E

$$P(X=k \mid E) = \frac{P(X=k \text{ and } E)}{P(E)} = \frac{P(X=k\,;\, k\geqslant 2)}{P(E)}$$

$$= \frac{\binom{10}{k}\left(\frac{1}{2}\right)^{10}}{1-\binom{10}{0}\left(\frac{1}{2}\right)^{10}-\binom{10}{1}\left(\frac{1}{2}\right)^{10}}$$

$$= \frac{\binom{10}{k}}{2^{10}-1-10} = \frac{\binom{10}{k}}{2^{10}-11}\ ;\ k\geqslant 2$$

You can leave the last answer like that or to be precise,

$$P(X=k \mid E) = \begin{cases} \dfrac{\binom{10}{k}}{2^{10}-11} & k\geqslant 2 \\ 0 & \text{otherwise} \end{cases}$$

Problem 7. A book has n typos. Two proofreaders, Sarah and Hannah, independently read the book. Sarah catches each typo with probability p_1 and misses it with probability $1 - p_1$, independently, and likewise for Hannah, who has probabilities p_2 of catching and $1 - p_2$ of missing each typo. Let X_1 be the number of typos caught by Sarah, X_2 be the number caught by Hannah, and X be the number caught by at least one of the two proofreaders.

(a) Find the distribution of X.

(b) For this part only, assume that $p_1 = p_2$. Find the conditional distribution of X_1 given that $X_1 + X_2 = t$

For typo i,

Sarah $\xrightarrow{p_1}$ catches the typo i

Sarah $\xrightarrow{1-p_1}$ misses the typo i

Hannah $\xrightarrow{p_2}$ catches the typo i

Hannah $\xrightarrow{1-p_2}$ misses the typo i

E_{is} = typo i is found by sarah

E_{iH} = typo i is found by Hannah

typo i is found by at least one of the two proof readers $= E_{is} \cup E_{iH}$

$$p = P(E_{is} \cup E_{iH}) = P(E_{is}) + P(E_{iH}) - P(E_{is} \cap E_{iH})$$

$$= p_1 + p_2 - p_1 p_2$$

X = number of typos that is found by at least one of the two proof readers

$$X \sim Bin(n, p)$$

$$P(X=k) = \binom{n}{k}(1-p_1-p_2+p_1p_2)^{n-k}(p_1+p_2-p_1p_2)^{k}$$

(b) X_1 = number of typoes found by Sarah

$X_1 \sim Bin(n, p_1)$

X_2 = number of typoes found by Hannah

$X_2 \sim Bin(n, p_2)$

since X_1 and X_2 are independent.

Method 1

$$P(X_1 + X_2 = k) = \sum_{i=0}^{k} P(X_1 = i)\, P(X_2 = k-i)$$

$$= \sum_{i=0}^{k} \binom{n}{i} p_1^{i} (1-p_1)^{n-i} \binom{n}{k-i} p_1^{k-i} (1-p_1)^{n-(k-i)}$$

$$= \sum_{i=0}^{k} \binom{n}{i}\binom{n}{k-i} p_1^{k} (1-p_1)^{n-i+n-(k-i)}$$

$$= p_1^{k} (1-p_1)^{2n-k} \underbrace{\sum_{i=0}^{k} \binom{n}{i}\binom{n}{k-i}}_{\binom{2n}{k}}$$

$$P(X_1+X_2=k) = p_1^{k} (1-p_1)^{2n-k} \binom{2n}{k} \implies X_1+X_2 \sim Bin(2n, p_1)$$

method 2

$$X_1 \sim Bin(n, p_1) \Rightarrow X_1 = I_1 + I_2 + \cdots + I_n$$

$$I_i \sim Bern(p_1)$$

$$X_2 \sim Bin(n, p_1) \Rightarrow X_2 = J_1 + J_2 + \cdots + J_n$$

$$J_i \sim Bern(p_1)$$

$$X_1 + X_2 = I_1 + I_2 + \cdots + I_n + J_1 + J_2 + \cdots + J_n$$

sum of $2n$ independent $Bern(p_1) \Rightarrow X_1 + X_2 \sim Bin(2n, p_1)$

$$P(X_1 = k \mid X_1 + X_2 = t) = \frac{P(X_1 = k \text{ and } X_1 + X_2 = t)}{P(X_1 + X_2 = t)}$$

$$= \frac{P(X_1 = k)\, P(X_2 = t-k)}{P(X_1 + X_2 = t)} = \frac{\binom{n}{k} p_1^{k} (1-p_1)^{n-k} \binom{n}{t-k} p_1^{t-k} (1-p_1)^{n-t+k}}{\binom{2n}{t} p_1^{t} (1-p_1)^{2n-t}}$$

$$P(X_1 = k \mid X_1 + X_2 = t) = \frac{\binom{n}{k}\binom{n}{t-k}}{\binom{2n}{t}} \qquad k = 0, 1, 2, \ldots, t$$

Problem 8. Suppose that the time T that it takes to complete a Homework in hours follows a distribution with density

$$f(t) = \frac{2(t^2 + t)}{27}, \qquad 0 \le t \le 3$$

What is the probability that a randomly chosen student finishes the exam during the second hour of the exam. That is, calculate $P(1 < T < 2)$.

$$P(1<T<2) = \int_1^2 f(t)\,dt = \int_1^2 \frac{2(t^2+t)}{27}\,dt$$

$$= \frac{2}{27}\left(\frac{t^3}{3} + \frac{t^2}{2}\right)\Big|_1^2 = \frac{2}{27}\left[\frac{8}{3} + 2 - \frac{1}{3} - \frac{1}{2}\right]$$

$$= \frac{2}{27}\left[\frac{7}{3} + \frac{3}{2}\right]$$

$$= \frac{2}{27}\left[\frac{14+9}{6}\right] = \frac{23}{81}$$

Problem 9. A factory produces high-quality glassware. When the furnace temperature is correctly maintained, the glass defect rate is 5%. However, there is a 10% chance that the temperature control system malfunctions, in which case the defect rate rises to 30%.

(a) Suppose a randomly selected glass item from the production line is found to be defective. What is the probability that the furnace temperature control had malfunctioned?

(b) A random sample of 20 glass items was inspected, and 4 of them were found to be defective. What is the probability that the furnace temperature control had malfunctioned?

M = tempreture control malfunction $\quad P(M) = 0.1$

D = defect

$P(D \mid M) = 0.3$

$P(D \mid M^c) = 0.05$

$$P(D) = P(D|M)P(M) + P(D|M^c)P(M^c) = (0.3)(0.1) + (0.05)(0.9)$$
$$= 0.075$$

(a) $$P(M|D) = \frac{P(D|M)P(M)}{P(D)} = \frac{P(D|M)\ P(M)}{0.075}$$

$$P(M|D) = \frac{(0.3)(0.1)}{0.075} = \frac{0.03}{0.075} = \frac{30}{75} = \frac{2}{5}$$

$$P(M|D) = \frac{2}{5}$$

(b) X = number of defectives

$$X \sim \text{Bin}(n = 20,\ P(D) = 0.075)$$

$$P(X=4) = \binom{20}{4} (P(D))^4 (1-P(D))^{16}$$

$$P(M \mid X=4) = \frac{P(X=4 \mid M)\ P(M)}{P(X=4)} = \frac{\binom{20}{4} (P(D \mid M))^4 (1-P(D \mid M))^{16}\ 0.1}{\binom{20}{4} (P(D))^4 (1-P(D))^{16}}$$

$$P(M \mid X=4) = \frac{(0.3)^4 (0.7)^{16} \cdot 0.1}{(0.075)^4 (0.925)^{16}} = \left(\frac{300}{75}\right)^4 \left(\frac{700}{925}\right)^{16} \frac{1}{10}$$

Problem 10. Each of $n \geq 2$ people puts his or her name on a slip of paper (no two have the same name). The slips of paper are shuffled in a hat, and then each person draws one (uniformly at random at each stage, without replacement). Find the mean and standard deviation of the number of people who draw their own names.

X = number of people who draw their own hat

$$X = I_1 + I_2 + \cdots + I_n$$

$$I_i = \begin{cases} 1 & \text{if a person } i \text{ draws their own hat} \\ 0 & \text{otherwise} \end{cases}$$

$$P(I_i = 1) = \frac{1}{n}$$

$$E[X] = \sum_{i=1}^{n} E[I_i] = n \cdot \frac{1}{n} = 1$$

$$\operatorname{Var}(X) = \operatorname{Cov}(X, X) = \operatorname{Cov}\left(\sum_{i=1}^{n} I_i , \sum_{j=1}^{n} I_j\right)$$

$$= \sum_{i=1}^{n} \operatorname{Var}(I_i) + \sum_{i \neq j} \operatorname{Cov}(I_i, I_j)$$

$$= \sum_{i=1}^{n} \operatorname{var}(I_i) + 2 \binom{n}{2} \operatorname{Cov}(I_1, I_2)$$

$$= n \cdot \frac{1}{n}\left(1 - \frac{1}{n}\right) + 2\binom{n}{2}\Big[E[I_1 I_2] - E[I_1]\, E[I_2] \Big]$$

$$I_1 I_2 = \begin{cases} 1 & \text{if } I_1 = 1 \text{ and } I_2 = 1 \\ 0 & \text{otherwise} \end{cases}$$

$$p(I_1=1, I_2=1) = \frac{1}{n}\,\frac{1}{n-1} = \frac{1}{n(n-1)}$$

$$\mathrm{var}(X) = 1-\frac{1}{n} + 2\binom{n}{2}\left(\frac{1}{n(n-1)} - \frac{1}{n^2}\right)$$

$$\mathrm{var}(X) = 1-\frac{1}{n} + 2\,\frac{n(n-1)}{2}\left[\frac{1}{n(n-1)} - \frac{1}{n^2}\right]$$

$$\mathrm{var}(X) = 1-\frac{1}{n} + 1 - \frac{n-1}{n} = \frac{2n-1-n+1}{n} = \frac{n}{n} = 1$$